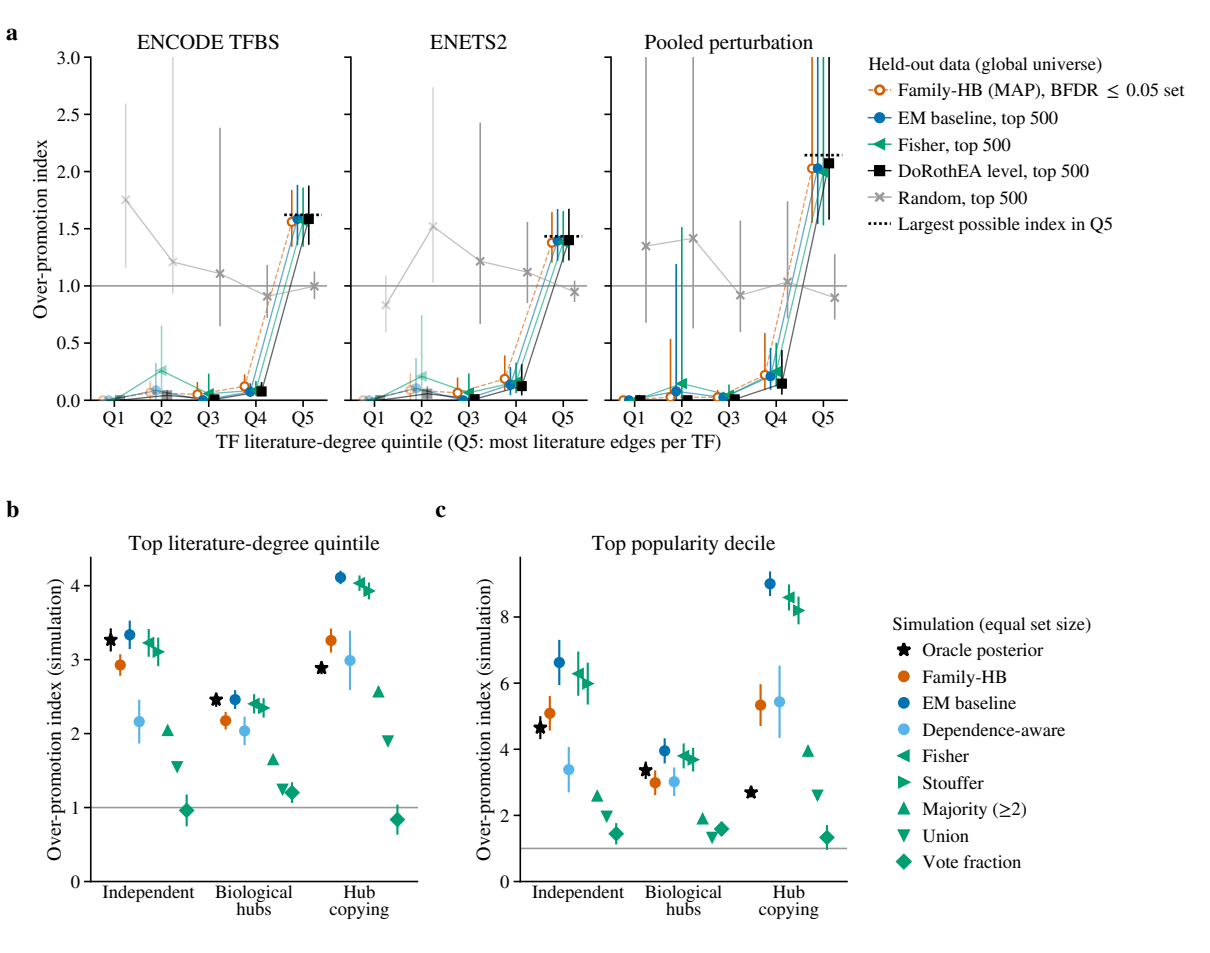
a
ENCODE TFBS
ENETS2
Pooled perturbation
Over-promotion index
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Q1
Q2
Q3
Q4
Q5
TF literature-degree quintile (Q5: most literature edges per TF)
Held-out data (global universe)
Family-HB (MAP), BFDR ≤ 0.05 set
EM baseline, top 500
Fisher, top 500
DoRothEA level, top 500
Random, top 500
Largest possible index in Q5
b
Top literature-degree quintile
Over-promotion index (simulation)
Independent
Biological hubs
Hub copying
c
Top popularity decile
Simulation (equal set size)
Oracle posterior
Family-HB
EM baseline
Dependence-aware
Fisher
Stouffer
Majority (≥2)
Union
Vote fraction